Chapter 6 — Common pitfalls in EDA

Even experienced analysts over-read charts, ignore quality problems, or stop after one pass

Learning objectives

- Recognize visualization overfitting, confirmation bias, neglected quality checks
- Premature closure of exploration

Overfitting with visualization

- Too many variables, categories, or colors clutter charts and invite false patterns
- Spurious correlations and cherry-picked time ranges can look convincing
- Prefer simple charts, fair axes, and follow-up checks

Misreading patterns

- A one-day sales spike may be a promotion, not a trend
- Weak correlations become overstated when charts are tuned to impress
- Example 6.5's weather cue still needs confirmatory analysis, not instant causation claims

Ignoring data quality issues

- Skipping missingness
- Section 6.4 exists to prevent exploring broken tables as if they were ground truth

Confirmation bias and lack of iteration

- Searching only for evidence that supports a favored hypothesis hides contradictory views
- EDA should be iterative: new plots after cleaning, new questions after surprises
- Example 6.4 warns against forcing linear stories on nonlinear data

Takeaways

- Simplify visuals, validate patterns, fix quality first, and iterate
- Pitfalls are cognitive as much as technical, process discipline reduces them

Next

- Complete the quiz for this part
- Continue to domain case studies